Chapter 11
University–Industry Relations in Clinical Trials

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11.1 Introduction

Clinical trials are essential for the development of new drugs, therapies, and medical devices to ensure safety and efficacy before they are approved for marketing and widespread public use. Randomized double-blind clinical trials are the gold standard for determining the most effective innovative therapies, devices, and procedures in healthcare. Whether evaluating an investigational drug, device, biologic, gene, or cell therapy, all new products require regulatory approvals, which typically demand scientific evidence of effectiveness through successful clinical trials.

The biopharmaceutical industry funds academic collaborators to plan, design, assess, and execute clinical studies. At many universities, clinical trials may be the largest source of funding for clinical research. Clinical studies are also important for the careers of clinician scientists. However, the actual value of clinical trials for investigational drugs, products, and devices is that they give patients access to innovative therapies through research studies. These trials compare new treatments to the current standard of care, ensuring that even patients in the control group receive the same high-quality care as nonparticipants. For hospitals and health systems, this is crucial because it enables them to offer cutting-edge therapies that are not yet widely available, enhancing their reputation and clinical enterprise. For patients seeking advanced treatment options, clinical trials offer a distinct advantage over traditional care by providing access to competitive, state-of-the-art therapies.

To ensure this advantage, institutional support for research administration, clinical trial operations (such as research coordinators and data managers), and critical resources like the investigational pharmacy, clinical trials office, and Institutional Review Board (IRB) are needed.

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11.1.1 R&D Pipeline

Essential to mutual success in medical advancements, universities and industries have collaborated through a variety of relationships to develop new therapies and study their effectiveness. These types of relationships can range from simple individual consulting agreements to complex multiparty, multistakeholder multiyear research collaborations. To address the complexities of human biology, chemistry, and physics of the human body, academia and industry depends on vital peer-reviewed publications, which serve as the essential performance indicator of academic excellence and noteworthy groundbreaking research.

As depicted in Fig. 11.1, the number of R&D pipeline products in phase I that are being evaluated in clinical trials has grown since 2012. Oncology remains the area of the biggest new clinical trial opportunities, amounting to 10.5% over the last 5 years.

Continuing the focus on oncology, Fig. 11.2 illustrates the clinical trial development pipeline for targeted small molecules and biotech therapies in oncology. This includes numerous new immuno-oncology treatments, precision medicine approaches, and next-generation biotherapeutics (such as cell, gene, and RNA therapies), which have seen significant growth in clinical trials for both rare and nonrare cancers over the past decade.

11.1.2 Types of Clinical Research

Table 11.1 represents the phases and types of clinical trials. Types include multi-centered, investigator-initiated, and sponsor-initiated. Trials may emerge from faculty discoveries. They may also be used for approved drugs.

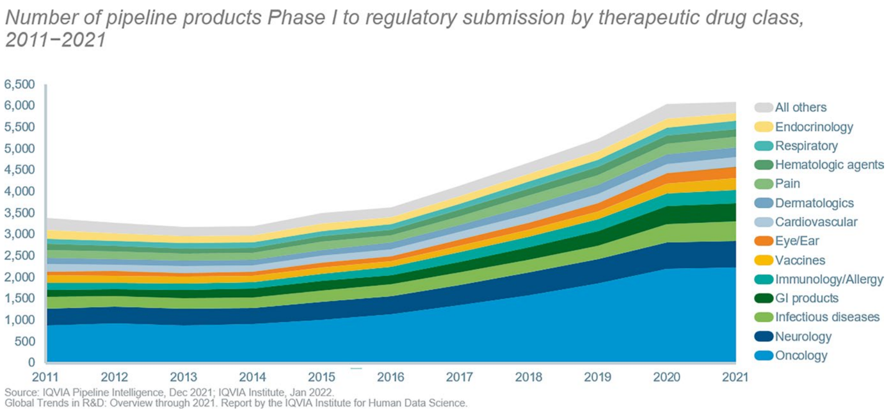

Fig. 11.1 Number of pipeline products produced in phase I submission by drug class

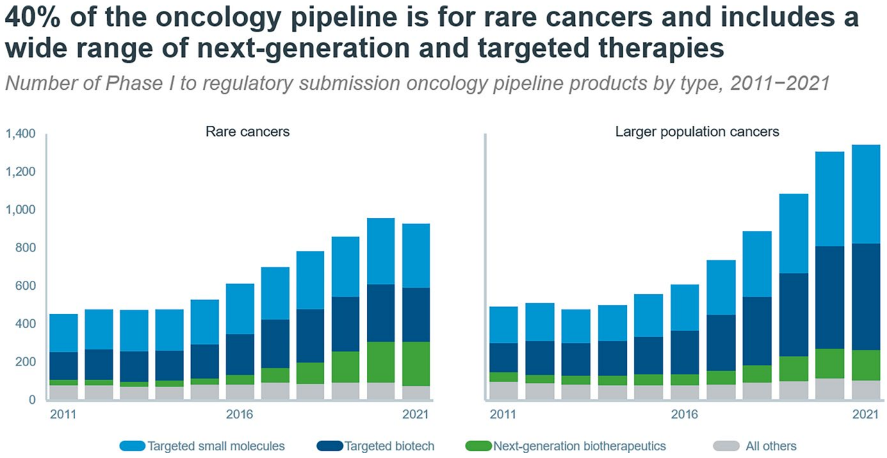

Fig. 11.2 Number of phase I oncology pipeline clinical trial regulatory submissions by class type and year

While most pharmaceutical preclinical studies are either conducted as work for hire through contract research organizations (CROs) or internally performed, there are exceptions where university faculty laboratories have proprietary cell, organoid, or animal models that the industry would like to use to test their therapies. The collaboration of preclinical and translational research between university faculty and industry sponsors paves the way for supporting data to inform questions being assessed in a human clinical trial. As another example, phase IV can lead to opportunities for publications and supplemental new drug applications (NDAs), providing opportunities for university faculty to determine new methods of use, more effective dosing, or other nuances of new products. These could render new supplemental indication approvals, provide possible safety warnings, or even the removal of approvals if the product is deemed unsafe for the broader public.

11.1.3 Steps in Beginning an Industry-Sponsored Clinical Trial

Investing resources to initiate, execute, and manage a clinical trial at a biopharmaceutical sponsor follows careful review and analysis of several factors. These include the trial's feasibility, expected return on investment, and probability of success—both in addressing the scientific objectives (e.g., achieving primary and secondary endpoints) and in ensuring smooth operational execution. Some trials, such as those involving new biologics, are inherently expensive and complex. This is due to challenges like scaling production for small, manufactured batches and the added intricacy of operationalizing first-in-human administration, closely monitoring

Table 11.1 Phases of clinical trials, key objectives, university and industry outcomes

Phase of clinical trials	Key objectives	Key activities	Industry and university outcomes
Phase O—-Exploratory Investigational New Drug (IND)	Preliminary data generation on investigational drug on human body and its organs/mechanism of action	Microdosing in a small number of volunteers to understand pharmacokinetics and pharmacodynamics in humans without expecting therapeutic or toxic effects (e.g., pre-dosing prior to tumor removal).	Inform decision-making about whether to proceed with further clinical development—measurable drug penetration into specific targeted organ.
Phase 1 (Safety and Dosing and Schedule)	Assessing safety, tolerability—acceptable toxic profile, pharmacokinetics, and pharmacodynamics of drugs	Conducted in small group (20-100) of healthy volunteers or patients with disease. Focus on identifying safe dosage range and side effects.	Industry—determination to proceed with a recommended phase 2 dose to determine optimal dosing for safety and efficacy.
Phase II (Efficacy and Side Effects)	To evaluate the efficacy of the drug and further assess its safety	Involves a larger group of patients (100-300) who have the condition the drug is intended to treat. Tests the effectiveness of the drug, optimal dose, and further evaluates short-term side effects.	Determine whether the drug works for the target condition and continue safety assessments.
Phase III Confirmatory Study compared to Standard of Care	Confirm drug's efficacy, monitor side effects, compare it to standard of care treatments, and collect information that will allow drug to be used safely in wider setting	Involves a larger group (1000–3000) of patients, often in multiple locations. The trial compares the new drug against the current standard treatment or placebo, often in a randomized and double-blind manner.	Provide comprehensive data on effectiveness, benefits, and the range of possible adverse reactions. Successful completion is required for regulatory approval.
Phase IV (Post-Market Surveillance)	Monitor the long-term effects after approval. This could be a regulatory obligation, especially if the product was granted accelerated regulatory approval	Conducted after the drug is on the market, involving a broad patient population. This phase aims to detect any rare or long-term adverse effects, identify any additional benefits, and refine understanding of the drug's risk–benefit profile.	Ensure ongoing safety and efficacy monitoring, possibly leading to adjustments in how the drug is used or additional warnings.

signs of adverse effects, including collecting and analyzing numerous pharmacokinetic (PK) and pharmacodynamic (PD) samples per patient.[4]

Universities can be approached in various ways to participate in an industry-sponsored clinical trial. Interest may be initiated by a contract research organization

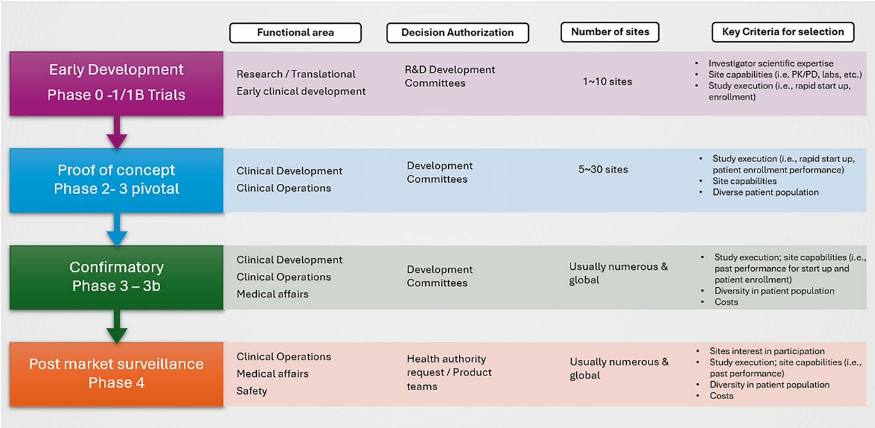

Fig. 11.3 Examples of industry-sponsored trial initiating functional areas, decision authorization processes, average number of sites, and criteria for site selection

(CRO), directly by the sponsor through different functional areas (such as translational research, clinical development, or medical affairs), or by the university itself. Universities may reach out to industry sponsors through channels like clinicaltrials.gov contacts, research and clinical teams, or medical affairs—specifically leveraging the role of medical science liaisons.

Figure 11.3 illustrates the financial and product funding decisions involved, key points of contact for initiating requests, and the sponsor's perspective on decision-making considerations.

11.2 Opportunities and Challenges

This section describes three key issues in conducting industry-sponsored clinical trials in universities: (1) budgeting, (2) speed of execution, (3) misperceptions, and (4) conflicts of interest. In each case, the issues differ from other types of industry-sponsored research.

11.2.1 Fair Market Value of Clinical Trial Budgets

Industry-sponsored clinical trials are often supported with a budget that pays for the cost of conducting research at fair market value. Within the budget, the industry sponsor reimburses the cost of patient care that is defined as research related (i.e., not standard of care), and reimburses institutional services, such as time and effort for administration, regulatory services, pharmacy, radiology, quality assurance, data management, and other ancillary services. Industry-developed budgets are required to be fair market value; that is, the industry shares actual negotiated budgets across different phases, therapeutic areas, and by geographic location of these trials to ensure that costs proposed are not inducements to participate or have undue influence on the institution. Biopharmaceutical companies subscribe to cost data-sharing services such as Grant Plan® and Grants Manager® to develop these fair market value budgets.

Hospitals and healthcare systems typically have predetermined rate cards for industry-sponsored clinical trials, which usually fall between commercial insurance rates and government payor rates. These costs can vary based on the institution's geographic location and demographic factors. Challenges arise when the fair market value estimated by the industry sponsor exceeds the trial budget allocated by the sponsor.

This issue is often seen at highly reputable academic institutions, where overhead rates are significantly higher, and upfront startup costs for preparing a clinical trial can be substantial, possibly leading sponsors to limit the number of academic institutions in multicenter trials. Community-based or decentralized sites offer competitive advantages due to their ability to start trials quickly, often using a centralized single IRB review, which accelerates start-up and enrollment. Additionally, international sites present further competition, as they can conduct trials at a lower cost sometimes as low as one-sixth the price, compared to academic medical centers in the United States (more information on international research can be found in Chap. 17).

Institutional overhead (indirect costs for industry-sponsored trials) varies from 20% to as high as 35% of direct costs. Unlike the federally negotiated facility and administrative (F&A) rate, industry sponsors cover administrative expenses as direct costs, including start-up activities such as IRB regulatory preparation. Study start-up costs at academic institutions can vary from mid-$20,000 to over $150,000 of nonrefundable administrative fees, which includes a long list of departmental reviews, such as radiology, pharmacy, therapeutic area protocol review, cancer protocol review, institutional IRB reliance administration fee, and central IRBs. Other review fees, such as Medicare Coverage Analysis and billing compliance fees, require a substantial investment of time and funds to ensure institutional performance in the study. Often, academic sites require greater than 120 days of start-up process, some extending to over a year, after the clinical trial has nearly completed enrollment.

11.2.2 Slow Start Challenges: The Role of Speed

The Clinical Trials Transformation Initiative (CTTI)-sponsored site metrics project aims to analyze start-up metrics by site type and IRB type. Lack of uniformity in clinical trial contracts and budgets, including inconsistent language and interpretations, creates delays, especially on contentious topics like indemnification and intellectual property. Factors such as varying infrastructure, limited resources, and excessive parties further hinder efficiency, while motivational factors, such as investigator-initiated or high-profile trials, can lead to compromises that leave university sites feeling underpaid or sponsors facing unexpected higher costs. These complex issues collectively slow the process of reaching timely budget and contract agreements.[3]

Significant opportunities exist to streamline the lengthy and complex timelines involved in study start-up activities, such as budget, contract negotiations, and the sequential review processes by multiple institutional committees, including the IRB and ancillary committees (protocol, departmental, radiology, pharmacy, etc.). These processes often create significant delays. Some institutions attempt to address institutional bottlenecks by outsourcing certain aspects of the clinical trial start-up management process, like Medicare Coverage Analysis, contract reviews, and IRB assessments, under the assumption that third-party providers can deliver a more centralized, faster process. However, industry sponsors report minimal benefits from these outsourced services. Instead, these providers often introduce additional administrative bottlenecks with limited accountability. As a result, study start-up times at some academic institutions frequently exceed six months.

The issue is complicated when sponsors outsource study start-up tasks to contract research organizations (CROs), which can further prolong timelines and restrict direct resolution between sponsors and universities. Budget and contract agreements are essential for managing financial liability, strengthening site-sponsor relationships, and ensuring regulatory compliance, though they can be time-consuming. Efforts to streamline these processes exist, such as contract templates from groups like Model Agreement Group Initiative (MAGI) and the Society of Clinical Research Sites' Common Language Evaluation and Reconciliation (CLEAR). Unfortunately, many sites still handle contract review through a legal reviewer, who often wordsmiths with inconsequential comments, such as changing promptly to immediately or changing the jurisdiction of state law. Additionally, contributing to the long timelines, the muti reviews are sequential, and budgets and contracts including payment terms, often are re-reviewed, resulting in delays.

Tools like contracts and grant management systems and document exchange portals help track progress and reduce duplication. However, the adoption of improvement initiatives faces challenges, including perceptions of templates being "too sponsor-friendly," disagreements over fair compensation, and limited funding or incentives for implementation. These barriers highlight the need for greater buy-in and resource allocation to drive efficiency in clinical trial setup with universities and industry sponsors.[4]

11.2.3 Misperceptions Overload

Like all complex relationships, collaborations between industry and academia are often hampered by misunderstandings regarding each other's intentions, motivations, and expectations. Intellectual property and publication rights are two areas where such partnerships frequently stall, bogged down by slow, bureaucratic processes involving multiple reviewers and intricate legal language. Universities tend to be wary of the profit-driven motives of industry, while sponsors often feel that academia seeks overly broad patent coverage and publication rights for the compounds, assets, and devices developed by industry partners.

To address these challenges, beginning with open and honest conversations about each party's needs and expectations would greatly improve the process. Such discussions can clarify intentions, reduce misperceptions, and help both sides find common ground. Unfortunately, legal, financial, and compliance professionals often operate in isolated silos without a clear understanding of the broader business or academic goals of clinical trials. This disconnect leads to unnecessary delays, frustration, and cost overruns, ultimately resulting in diminishing returns for sponsors, researchers, and, most importantly, patients.

Clinical trials require multiple layers of institutional, financial, ethical, and compliance approvals. In a university medical center setting, this complex process can be excruciatingly long, averaging greater than six months to upward of 18 months. Some institutions outsource these processes, such as Medicare billing compliance review to centralized contract organizations that promise efficiency. However, the most efficient processes involve dedicated institutional staff that shepherd the clinical study through every step of the complicated list of requirements. Just as start-up timelines can be lengthy, the administrative costs for these reviews can also be immense. At academic medical centers, the total administrative charges and overhead fees for start-up average $25,000 to upward of $200,000 before a single patient is enrolled.

Many other factors affect industry-sponsored clinical trials, including the following.

Conflicts of Interest: Managing payments that go directly from sponsor to faculty, such as speaker fees. Also, startup-related conflicts and institutional conflicts from university-owned patents.

Sunshine Act: Reporting requirements, enacted as part of the Affordable Care Act, require pharmaceutical and medical device companies to report payments made to physicians and teaching hospitals. This includes payments related to clinical trials, such as consulting fees, research funding, and travel expenses. Payments can be viewed through the Open Payments database. The law aims to increase transparency and prevent conflicts of interest by making this information publicly accessible. While clinical trial payments are disclosed, there are certain research-related payments that may qualify for delayed publications to protect confidential information before a study is published.

Publications: Unique issues for multicenter trials. Timing of publications. Preregistration of trials. Publication of both negative and positive results.

Organization: Different models for both universities and industry for how the clinical research operation is organized and coordinated, including regulatory. Function of a centralized clinical trial office versus decentralized. Role of university versus medical school. Role of CROs in the process has its advantages and disadvantages.

Institutional Conflicts of Interests Institutional conflicts of interest arise when a university stands to gain financially from the success of an investigational therapy undergoing clinical trials. This is especially concerning when the university holds patents or stands to benefit financially if the therapy is commercialized. As clinical trials advance toward market approval, universities should carefully evaluate and actively manage these conflicts to ensure integrity in the research process. Academic institutions have a long-standing collaborative and productive relationship with the biopharmaceutical industry where conflicts of interest are disclosed and actively managed. These institutions recognize the importance of industry collaboration and actively engage industry sponsors in identifying ways to align ways of working to serve the best interests of patients and the public.

Inter-Institutional Organization: Due to the academic research obligations of faculty, the structure of medical centers and medical schools, and their intersection with clinical practice, institutions may have varying mechanisms for financial and institutional oversight in clinical trials. Clinical trials regulated by the FDA under the “1572 – Statement of the Investigator” require identification of the principal investigator, sub-investigators, and facilities where study activities and clinical data will be generated. Section #3 of the 1572 captures the locations of study procedures, such as medical centers, pharmacies, storage, or tissue collection sites where investigational products and clinical functions are carried out (Fig. 11.4).

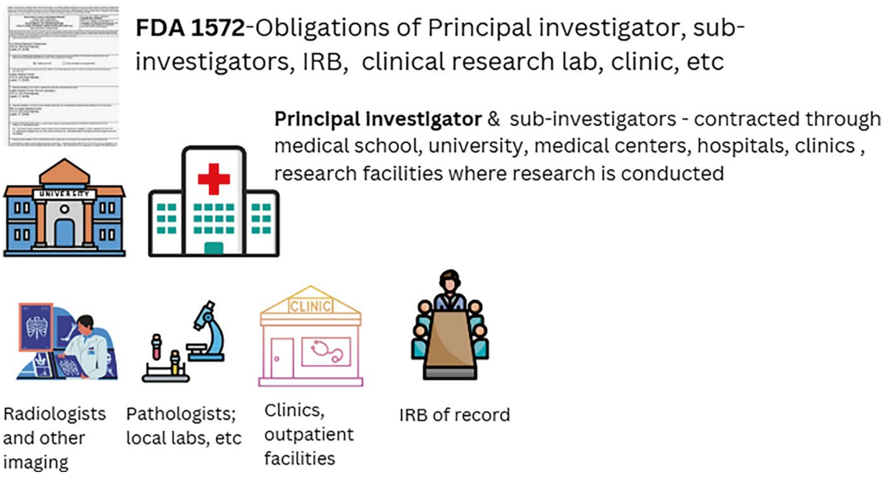

Fig. 11.4 1572 form outlines obligations for principal investigators and IRBs involved in clinical trials of investigational drugs and biological products under 21 CFR part 312 (Investigational New Drug application (IND) regulations) and describes how to complete the Statement of Investigator (Form FDA 1572)

11.3 Options, Solutions, and Impact

11.3.1 Accelerating Trials

Over the past two decades, industry and universities have pursued a concerted effort to streamline and improve clinical research by centralizing administrative efforts for industry-funded trials. The multistakeholder, public–private partnership of the Clinical Trials Transformation Initiative (CTTI)[2] (Alexander, 2018; Tenaerts, 2018) developed and drove adoption of practices that increased the quality and efficiency of clinical trials. The partnership identified strategic focus areas to eliminate barriers to operational efficiencies, improve trial quality, address patients as specific stakeholder engagement, and address public health concerns. CTTI led and issued more than 30 sets of evidence-based recommendations, frameworks, and tools to better clinical trials.[3]

Efforts to streamline industry clinical trials continue to face challenges inherent in complex human systems. Multiple committee reviews and administrative processes often delay study start-up and drive-up costs. Cost-effective trial sites at US academic medical centers and universities are becoming increasingly rare. However, highly engaged principal investigators and dedicated research support teams can significantly accelerate the process, achieving study start-up and activation in as little as 60 days. Institutions with specialized regulatory teams, budget and contract experts, and compliance staff work in parallel to facilitate simultaneous review and approval. Many also utilize central or single IRB reliance agreements, allowing institutional modifications to be efficiently implemented upon receipt of regulatory documents. Advanced regulatory management tools, such as Florence™ eBinders, further streamline the process by digitizing, automating, and integrating investigator site files, participant binders, and logs while enabling remote monitoring for sponsors and CROs.

Key clinical trial study activation metrics, such as those in Table 11.2, help assess the efficiency of site activation and identify bottlenecks. These metrics are critical for academic medical centers, research institutions, and sponsors to streamline processes and improve study timelines.

Institutions that optimize these metrics can significantly improve study startup times and improve operations, making them more attractive to industry sponsors and CROs. These institutions have intentionally streamlined their processes to reduce timelines, enhance efficiency, and accelerate study activation. They implement clear protocols for parallel reviews, expedite IRB approvals through reliance agreements, and leverage technology to provide industry sponsors with transparency on clinical trial activation requirements. Frequent communication with sponsors and CROs regarding site activation requirements is also critical to ensure a smooth and efficient process.

To further strengthen their appeal, these institutions should proactively track key performance indicators, such as those listed previously in Table 11.2: the cycle time for receipt, approval, and signing of regulatory packets and contracts, as well as

Table 11.2 Clinical trial study start-up metrics comparison

Study startup performance indicator	Description	Nonacademic site cycle times	Typical academic cycle times
Time from site selection to site activation	Measures the total cycle time from when a site is chosen to when it is fully activated.	> 45–90 days.	> 120 days.
Number of committee reviews	Number of various committee reviews and if approvals must be sequential or can be done in parallel.	0-1.	~3 committees.
Time to regulatory packet submission	Number of days until site submits required regulatory documents after selection.	0 ~>30 days. Usually can use central IRB (sponsor selected).	>30~45 days. Sites usually need to customize ICFs and submit IRB reliance agreements to local IRB.
Regulatory submission to IRB approval	Evaluates IRB review efficiency, especially when using central IRB reliance agreements, understanding institutional-specific requirements of local review.	0 ~>15 days. Independent commercial IRBs generally have multiple committees/meetings in a month.	> 30–60 days. Depends on timing of IRB committee meetings and submission deadlines.
Receipt of contract/budget to contract/budget execution	Measures the total time for contract, budget, and financial attachments, such as payment terms are reviewed and approved. Total time is inclusive time for final signatures.	>30–45 days.	>60–120 days. May include additional reviews of Medicare Coverage Analysis; billing plan, etc.
Time from contract execution to site activation	Measures the final steps before the site is ready to enroll patients.	>30–45 days.	> 30–60 days. May require additional AMC activation requirements for ready for enrollment.

study activation and first patient consent dates. Utilizing these metrics can help attract new clinical trials and new industry sponsors. Achieving study activation in under 60 days is exceptional for an academic medical center, and institutions that can consistently meet this benchmark will draw significant interest from industry sponsors. Industry sponsors capture the following site clinical trial enrollment performance metrics to gauge operational efficiencies and site performance (Table 11.3).

These metrics are critical for evaluating the efficiency, accuracy, and overall performance of clinical trial sites, regardless of setting (academic or nonacademic). Industry sponsors place significant value in timeliness of data entry, data accuracy, and quality.

Table 11.3 Operational and enrollment metrics

Clinical trial performance metrics	Measurement
Time from site activation to first patient consented	Assesses how quickly the site can begin enrolling participants after activation.
Time from first patient consented to first patient dosed	Measures the site's ability to initiate investigational treatment after patient consent. This is usually protocol specific, as some clinical trials require extensive screening procedures, including central genomic labs or pathology confirmation
Time from study visit completion to completion of electronic data capture (EDC) entry	Measures the time elapsed between a patient's visit and the entry of corresponding data into the electronic data capture system. Faster data entry ensures timely monitoring, reduces delays in data analysis, and helps maintain data integrity.
Number of queries	Total number of queries raised by study team: data managers, monitors, or automated systems due to inconsistencies, missing data, or errors in the electronic case report form (eCRF). A high number of queries may indicate data quality issues, training gaps, or system inefficiencies.
Number of days to query resolutions	Metric tracks the time taken to resolve data queries from the moment they are raised until they are addressed and closed. Faster query resolution improves data quality, ensures regulatory compliance, and prevents delays in study milestones.

11.3.2 Master Protocols

Master protocols are an innovative, collaborative approach to clinical trials that benefit universities, industry, and patients by making research faster, more cost-effective, and more adaptable to evolving scientific knowledge. They provide overarching clinical trial study designs that allow multiple research questions to be answered within a single-trial structure, streamlining research processes and facilitating more efficient drug development. Unlike traditional trials that typically focus on one treatment or disease subgroup at a time, master protocols can test multiple therapies, diseases, or specific patient populations within the same framework. They are particularly valuable in complex areas like oncology, infectious diseases, and personalized medicine driven by genomics, biomarkers, and other prognostic indicators that predict responses based on basic or translational science.

Biopharma sponsors struggle to execute clinical trials in a cost-efficient manner due to a variety of reasons like slow start-up, recruitment, enrollment, and retention challenges of patients, and lack of qualified or interested investigators. Master protocols provide an opportunity to address specific patient populations and combinations of therapeutic agents, as well as explore expanded opportunities for drugs to target different diseases. Master protocols are usually a collaboration between a collection of aligned goal-oriented stakeholders across disease and targeted patient population, which provides a clear path toward expediting treatments to patients. There are three main types of master protocols:

- **Basket Trials**: These test the effects of a single treatment on multiple diseases or subgroups with a common characteristic (e.g., a specific genetic mutation). Several industry sponsors have such trials available based on specific genetic or molecular markers, some well-known examples include *MyPathwayTrial—Genentech's* targeted therapies with patients with advanced cancers with specific molecular alterations, *KeyNote-158*—Merck's pivotal basket trial for patients with microsatellite instability-high (MSI-H) cancers that lead to pembrolizumab's tissue-agnostic FDA approval.
- **Umbrella Trials:** These evaluate multiple treatments for a single disease, usually by assigning patients to different substudies within the larger trial based on biomarkers or other disease characteristics. A successful example is LLS's Beat AML where the master clinical trial protocol allows different patients, depending on the genetic features of their AML, to enroll in Beat AML substudies using different treatment approaches. This model includes 11 academic medical center sites, 12 industry sponsors, and 4 genomic labs. Advanced technology provides genomic analysis to match patients to precision treatment within days.
- **Platform Trials:** These are adaptive trials that allow for the continuous addition of new treatments or study arms as the trial progresses. Treatments can be added or removed based on real-time results, allowing for a more flexible and ongoing investigation.

Master protocols bring several benefits to both industry (pharmaceutical companies and biotech firms) and academic research institutions:

- **Efficiency and Cost-Effectiveness**: By integrating multiple research objectives within one protocol, master protocols save time and resources. Instead of conducting separate trials, researchers and companies can streamline testing, which reduces costs and accelerates drug development.
- **Faster Patient Enrollment:** Since master protocols are often larger and more flexible in design, they offer more opportunities for patient participation. This is especially valuable in rare diseases, where finding eligible patients can be challenging.
- **Adaptive Design and Flexibility:** These protocols can adjust as the study progresses, allowing new treatments therapies to be added and ineffective ones to be dropped. Platform trials enable continuous evaluation of new treatments without launching separate trials. This flexibility aligns with the dynamic nature of scientific research and enables a more responsive approach. Adaptive trial features allow modifications based on interim results, making trials more responsive.
- **Collaboration Among Advocacy Groups, Academia, and Industry**: Master protocols are well-suited for partnerships as they bring together multiple stakeholders with shared goals. One of the most important stakeholder types is patient advocacy groups, which have an element of mission-driven research funding support. Such groups, like Leukemia & Lymphoma Society (LLS), PANCAN, American Cancer Society, provide financial support to bring together universities and industry sponsors to collaborate across different modality of treatments. Universities benefit from access to industry funding, resources, and data, while

companies gain access to academic expertise and patient populations. Industry sponsors would need to have data sharing agreements with other sponsors to share specific information about the combination profile of their therapies.

- **Accelerated Research and Innovation:** The adaptable nature of master protocols speeds up research timelines, which is especially important in areas with unmet medical needs. Faster trials mean quicker access to new therapies, benefiting patients and bringing innovative treatments in healthcare systems.
- Regulatory agencies, including the FDA, support master protocols because they offer a clearer and more cohesive approach to assessing multiple treatments, potentially simplifying the approval process.

Master protocols have transformed clinical trial design, benefiting universities by expanding research capabilities and industry by optimizing drug development, ultimately leading to faster and more effective medical advancements.

11.3.3 Other Best Practices

There are many examples of university–industry best practices. Most centers around specific disease/therapeutic areas, such as cancer, neurology, cardiology, or immunology. These best practices emerged as the growing investigational therapies needed partnerships with academic institutions and the approach from each party to work in collaboration to achieve common goals.

Key strategies include early communication, clear goal alignment, mutual trust building, designated champions within each organization, streamlined processes for collaboration, joint research planning, open data sharing, addressing intellectual property concerns upfront, and fostering a culture of shared learning and feedback. In sum, strategies should ensure both sides understand each other's needs, motivations, and timelines while prioritizing patient well-being and ethical research practices throughout the partnership.

11.4 Takeaways and Future Strategies

Modernization of clinical trials is necessary to transform every stage of the research process, from scientific study design to operational execution, with the integration of advanced technologies such as artificial intelligence (AI) and real-world evidence (RWE). AI-driven analytics help improve site selection, patient recruitment, trial monitoring, and data analysis. RWE, sourced from electronic health records, wearable devices, and patient registries, provides valuable insights into treatment effectiveness in diverse populations, complementing traditional clinical trial data. Institutions that embrace technological advancements to enhance their performance and leverage vast healthcare data will remain valuable partners to the industry.

Collaboration between industry and universities plays a pivotal role in driving these innovations. Academic institutions contribute cutting-edge research, methodological advancements, and access to diverse patient populations, while industry partners provide funding, regulatory expertise, and large-scale implementation capabilities. These partnerships facilitate adaptive trial designs, decentralized trial models, and faster regulatory approvals, ultimately accelerating the development and delivery of new therapies to patients. As the field continues to evolve, fostering strong UI relationships will be essential in shaping the future of clinical research and improving healthcare outcomes globally.

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